

Assessing the macro-environmental factors for waste-to-energy in Brazil: A comparative study with the USA, China, the EU, and India using PESTEL and panel data analysis

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ARTICLE INFO

Keywords:

Environmental legislation
Waste-to-energy innovation
Sustainability
Resource management
Data analysis

ABSTRACT

This study investigates the macro-environmental factors influencing the Waste-to-Energy (WtE) sector in Brazil, offering a comparative analysis with the USA, China, the EU, and India. Recognising WtE's role within the circular economy, the research addresses Brazil's unique challenges in urban waste utilisation despite its significant renewable energy profile. Employing a mixed-method approach, including PESTEL and panel data analysis, the study assesses Political, Economic, Social, Technological, Environmental, and Legal factors across these regions to derive actionable public policy recommendations. Key findings reveal that Brazil, similar to emerging economies like India and China, possesses the potential to adopt favourable WtE policies irrespective of its economic status. While Brazil demonstrates a competitive advantage in agricultural waste valorisation, the study highlights a notable underdevelopment of biogas and biomethane strategies from municipal waste compared to international counterparts. The paper concludes by proposing specific public policy enhancements for Brazil, drawing vital insights from the compared nations' successful experiences and diverse landscapes to foster sustainable WtE development.

1. Introduction

The United Nations' twelfth Sustainable Development Goal (SDG 12) calls explicitly for establishing economic frameworks that foster sustainable consumption and production patterns. In direct opposition to the traditional linear model of "take-make-dispose," the circular economy paradigm offers a compelling alternative. This model, often complemented by Waste-to-Energy (WtE) strategies, champions a holistic approach to waste reduction through diverse techniques such as reuse, repair, retrofitting, and recycling [1,2]. This paper intends to demonstrate that by examining international best practices in WtE from major developed and developing countries and regions, Brazilian policymakers can leverage significant economic, social, and technological

opportunities to enhance the WtE industry in the country through various productive routes.

In this article, we focus on three key WtE technologies that allow for a robust comparison across the diverse realities of the selected countries. Firstly, municipal solid waste (MSW) combustion in WtE plants involves incinerating MSW to generate energy, typically electricity or heat. This method represents a primary industrial approach for converting waste materials into useable energy, often in facilities capable of processing post-recycling waste streams. Secondly, methane capture at MSW landfills utilises technology to collect methane gas produced during the decomposition of organic materials. This captured methane can then be repurposed as a renewable energy source for electricity generation or various other applications, effectively mitigating greenhouse gas

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<https://doi.org/10.1016/j.renene.2025.124311>

Received 12 October 2024; Received in revised form 26 August 2025; Accepted 26 August 2025

Available online 3 September 2025

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emissions from landfills. Finally, anaerobic digestion of source-separated organics involves the biological conversion of organic waste into biogas, primarily methane, through microbial processes without oxygen. This technique is applied to source-separated organic waste from various streams, which may include mixed bio-waste from municipal sources. Korhonen et al. [3] argue that the main challenge for sustainable development lies in the linear flow of materials and energy between nature and the human economy, where inputs move unidirectionally through the production system and are discarded as waste [4–7]. Achieving these goals requires public policies to transform waste into energy.

Data and literature attest that Brazil grapples with a dual reality. On the one hand, it ranks among the twenty largest economies with the highest rates of renewable energy in both its electricity and energy matrices [8,9]. On the other hand, it is a country with significant production of untreated waste and limited utilisation of urban waste for energy generation [10]. However, Brazil excels in using agricultural waste for energy production [11]. WtE is treated within the circular economy context. In a circular economy, outputs serve as inputs in the productive system [12,13], closing the loop in a material and energy cyclical flow [14,15]. This concept aims to maximise the use of the energy and economic value embedded in natural resources [15]. Besides environmental benefits, the linear economy neglects the economic potential of discarded products, which could promote jobs and industries [3]. A circular economy approach could enhance urban solid waste management (USWM) in the face of growing global waste production, which is projected to rise from 1.3 billion to 2.2 billion tons by 2025 [16]. Much of this waste is in landfills and open dumps, causing negative environmental and health impacts [17,18]. Effective waste management prioritises reuse, repair, remanufacturing, and recycling to reduce natural resource extraction [2]. Maximising resource energy and economic value is essential for categorising Urban Solid Waste Management (USWM) within a circular economy project. The WtE approach can encompass various technologies: from the combustion of municipal solid waste (MSW) in WtE plants, production of biogas through organic anaerobic decomposition, which can be used in industry, transportation, and electricity generation, to the capture of methane at MSW landfills [19,20]. This approach reduces pollution and natural resource extraction while converting waste into an economic good [21,22] and prevents methane emissions from untreated waste [23]. The broad consideration of the term WtE in the article allows for comprehensive comparisons of the realities across the selected regions.

There is a notable gap in the literature regarding using available statistical data in panel data analysis to consider the Brazilian scenario compared to other countries practising effective WtE strategies. Although there are other studies based on the PESTEL methodology for strategic analysis of the energy sector [24,25], the Brazilian case, compared with other economies, is lacking. The PESTEL framework, which stands for Political, Economic, Social, Technological, Environmental, and Legal factors, is an essential analytical tool used throughout this study to evaluate external macroenvironmental influences on sector strategies. More studies are needed to compare strategies and macro-environments of the European Union (EU), the United States (USA), China, and India. Therefore, this article presents three innovative points for considering Brazilian public policies for WtE: a study based on panel data, a comparison of the Brazilian case with other major economies, and a strategic analysis of this industry using the PESTEL framework. The environmental scanning analysis can better offer public policy recommendations to develop the WtE sector in developing countries by providing a framework grounded in robust comparative data with other economies [26,27].

This article provides a comprehensive strategic management assessment of Brazil's WtE industry. We'll conduct a rigorous comparative analysis with established and rapidly developing WtE sectors in the European Union (EU), the United States (USA), China, and India. Our goal is to formulate tailored public policy recommendations for Brazil's

unique context and challenges.

To achieve this aim, we address the following research questions: RQ1: How do the PESTEL factors for WtE in Brazil compare with those in the USA, China, the EU, and India, based on quantitative data? RQ2: How can public policies in each region be evaluated based on their performance in WtE development? RQ3: What specific public policies could Brazil adopt to advance its WtE initiatives, drawing actionable insights from the experiences of these other regions or countries?

Our key findings highlight that Brazil's status as an upper-middle-income economy does not hinder the adoption of more supportive public policies for WtE, underscored by the successes observed in India and China. Brazil has already demonstrated strong capabilities in the agricultural sector, where it competes effectively with the EU and USA in utilising animal fat and sugarcane waste for electricity and biofuel production. Furthermore, Brazil's higher per capita methane emission rates can be significantly lowered by implementing best practices and targeted strategies compared to the other nations in this study. Initial results from policies promoting biogas and biomethane use are auspicious.

The country could also integrate WtE within the urban context, leveraging its positive experience in the biomass industry [19,20]. However, the integration within the urban context is not widely implemented in Brazilian municipalities compared to the strategies employed in the E.U., USA, China, and India.

This article has certain limitations. Our in-depth focus on the legal and political aspects primarily centred on Brazil and the European Union. Additionally, our consideration of the Circular Economy was explicitly confined to the WtE context. Our literature review was limited to articles indexed in the Scopus platform. However, a systematic review methodology was employed for screening due to the high volume of relevant articles. The decision to restrict our search to articles published in English allowed us to prioritise publications from internationally recognised journals. Future research will involve applying machine learning and advanced statistical techniques to gain a deeper understanding of key factors for panel data analysis, with a view to extending comparisons to other Latin American countries. Further deepening of future research will also include evaluating patents within the WtE industry and considering the crucial role of recycling critical raw materials for the energy transition.

In this context, the present article is structured into five sections, beginning with this introduction. Section 2 details our methodology using a mixed approach combining PICO, PRISMA, PESTEL, and Data Panel methods. In section 3, we present the results of our literature review analysis. Section 4 uses the PESTEL framework to compare Brazilian WtE policies with those of the USA, EU, India, and China. Finally, section 5 offers a discussion on the Brazilian case. The conclusion section follows at the end.

2. Materials and methods

2.1. Literature review using PICO-PRISMA frameworks

The methodology includes a literature review and subsequent qualitative analyses to reach the proposed objectives. Additionally, a study will be conducted to explore the potential lessons that can be learned from Brazilian and EU legislation. Finally, the Brazilian Renovabio Programme has been chosen as a case study due to its success in promoting bio-energy technologies to achieve the country's renewable energy targets.

The methodology used in this paper follows the steps highlighted by recent studies [28–33] to produce a reliable literature review, and Fig. 1 summarises this article's procedures. We limited the articles to those in the Scopus database that were written in English. We justified that this strategy is widely employed in academic papers in high-standard journals. Furthermore, the availability of the data enables the identification of trends in the bibliography more effectively. Limiting the language of

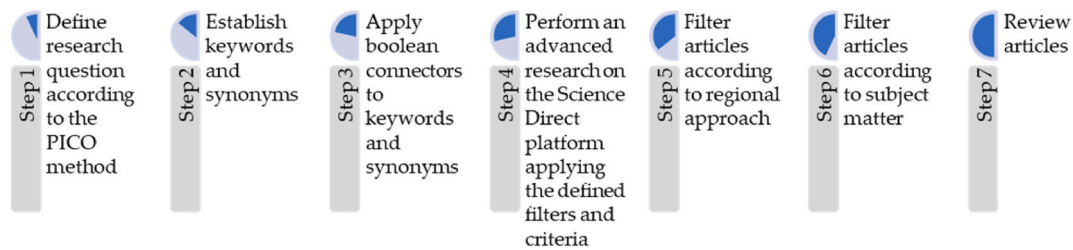


Fig. 1. PICO-PRISMA Methodology summary.
Source: Elaborated by authors based on [28].

the articles to English also facilitates access to publications in journals with greater international recognition.

As shown in Fig. 1, the first step determines the research question according to the PICO-PRISMA methods[51]. The PICO anagram presents essential points of the research question, in which “P” stands for population or problem, “I” stands for intervention, “C” stands for comparison, and “O” stands for the outcome [28]. To achieve this article’s objective, the question to be answered is “What public policies can be adopted in Brazil to develop WtE?”. In this case, the problem is “urban solid waste management and WtE”; the intervention is “legislation”; the comparison is “Brazil”; and the outcome is “recommended policies”.

Accordingly, the proposed keywords with Boolean connectors were WtE, biogas, urban solid waste management, circular economy, legislation, policy, Brazil, and the selected countries [28].

Panel data methodology, or longitudinal data analysis, involves examining multi-dimensional data with measurements taken over time. This paper considers data from 2000 to 2022, with some exceptions due to unavailability. This approach is advantageous because it allows researchers to control for individual heterogeneity, provide more informative data, and better capture dynamics over time. Panel data analysis can improve the efficiency of statistical estimations and enable the detection and adjustment of measurement errors and omitted variables. It facilitates a deeper understanding of temporal effects and causal relationships in empirical research by analysing variations within and between subjects over a time series [34].

Market data were the criteria for comparing Brazil and the E.U., the United States, India, and China cases. Considering its participation in world production, Brazil is leading in Installed Capacity from Bagasse and its relevance in electrical energy from biofuels and renewable waste. The E.U. leads in producing biogas, solid biofuels, renewable waste, liquid biofuels, bioenergy, and other solid biofuels relevant to renewable

municipal waste and biofuels. European experience can help strengthen the Brazilian biogas, bioenergy, and renewable municipal waste industry, which is underrepresented in the country (Fig. 2). Brazil’s productive potential can be seen because it has the fifth-largest urban population globally and produces urban waste (Fig. 3).

The chosen platform was the ScienceDirect website, which is widely used by international scientists in the energy field. In addition, the period filter was applied to select articles between 2018 and 2022, and only “Review articles” and “Research articles” were analysed. As a result, 203 articles were found (Figs. 4 and 5).

We filtered the articles according to their regional approach in the next step. Because this study aims to raise lessons from the Brazilian and EU experiences, all the articles that examined countries and regions outside these two were rejected. Fig. 2 shows the results from this stage.

Furthermore, a second filter was applied based on the article’s subject matter. Articles approaching one or more topics analysed here were approved at this stage. Fig. 3 details the criteria used, as well as the results found.

In a systematic review of the literature, the analysis commenced with 169 articles, leading to the rejection of 66 articles, primarily for not addressing urban waste. Notably, 68 % of these rejected articles did not deal with urban waste. Other specific exclusions included articles on industrial waste (24 %), food production waste (15 %), and smaller percentages focused on fuel production waste, nuclear waste, and other sectors such as marine, mining, chemical, energy, construction, and transportation. Of the 103 approved articles, 4 % explored the intersections of urban waste, energy, and circular economy, while 8 % addressed waste and circular economy. A further 2 % examined the relationship between the circular economy and energy. Articles on urban solid waste management comprised 14 %, and those focusing solely on urban waste accounted for 31 %. This comprehensive

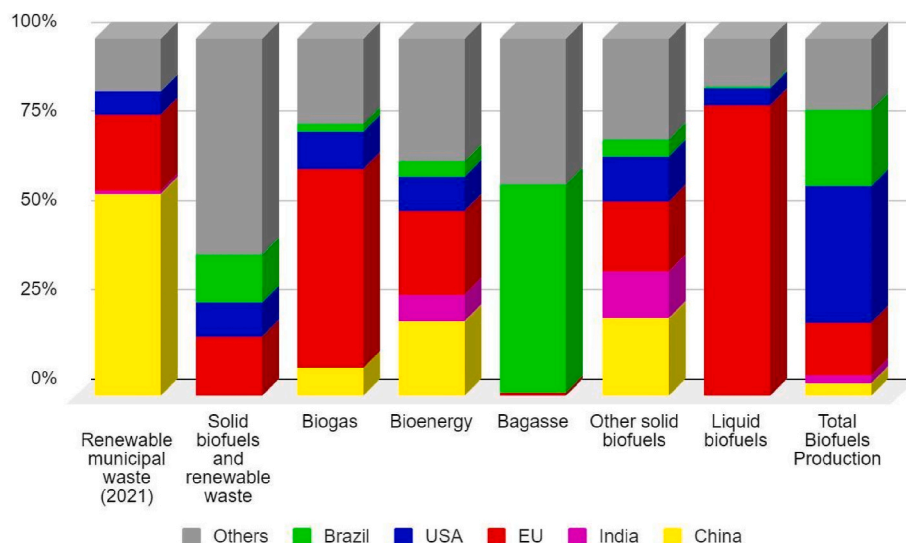


Fig. 2. WtE technologies power installed (MW) and total biofuels production (TWh). Source [35].

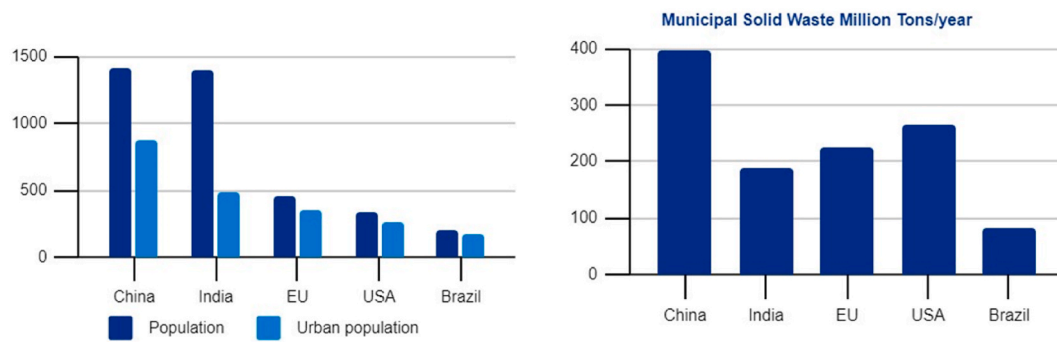


Fig. 3. Population, Urban Population and Municipal Waste Million Tons/year. Source > Urban population share. Source: [36].

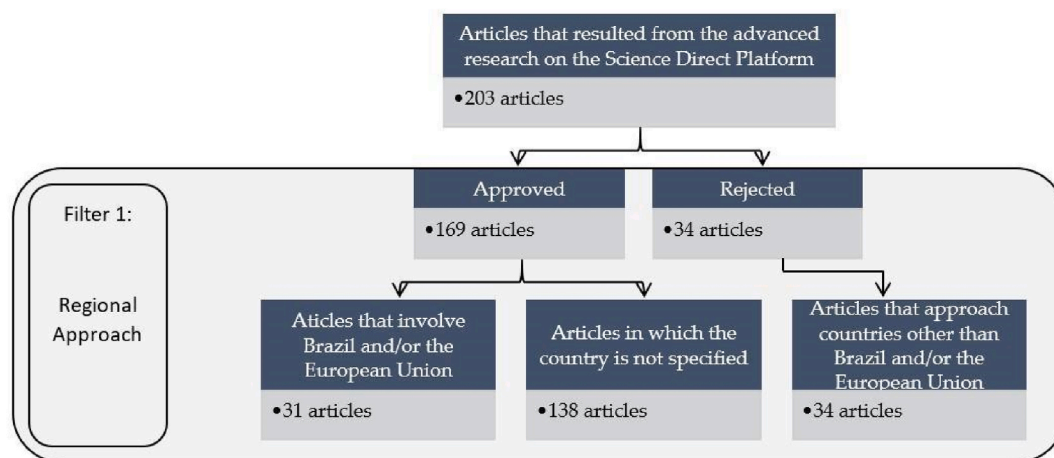


Fig. 4. Filter 1 details and results. Elaborated by the authors.

assessment highlights the thematic distribution of articles addressing urban waste and sustainable paradigms within the reviewed literature.

The PESTEL model is a strategic framework used to analyse the external macro-environmental factors with potential impact within an organisation, country, or industry [24]. Each aspect of the PESTEL model encompasses a distinct set of influences. **Political** factors pertain to the impact of government policies, regulations, and political stability on business operations. **Economic** factors involve the broader economy's condition, including inflation, interest rates, economic growth, and exchange rates, which can affect an organisation's performance and decision-making. **Social** factors consider societal trends and shifts, such as demographic changes, cultural norms, and consumer attitudes, which can influence market demand and organisational practices. **Technological** factors encompass innovations, technological advancements, and the rate of technological change, affecting how companies produce goods and services and interact with their markets. **Environmental** factors address ecological and environmental aspects, including sustainability practices, environmental regulations, and the impact of climate change, which are increasingly significant for business strategy. **Legal** factors refer to the legal environment in which an organisation operates, including laws related to employment, health and safety, competition, and intellectual property, which can dictate operational compliance and legal obligations.

3. Literature review in waste-to-energy and circular economy

To meet current needs while safeguarding resources for future generations, maximising the efficient use of natural resources is paramount [37]. However, the prevailing linear economic model, characterised by an extract-transform-use-discard sequence, has led to irreversible

environmental impacts, challenging its ability to sustain future generations [38]. Environmentally, this system has resulted in uncontrolled waste disposal, posing threats to land and marine ecosystems [39]. Moreover, from a value-maximisation perspective, the linear model leads to energy dissipation during production, with materials often discarded within six months of manufacturing [5] (p. 2).

In contrast, the circular economy aims to 'close the loop' within production systems, where end products become input materials for subsequent cycles [15]. This loop cannot be closed perfectly and indefinitely since some material waste and energy will always be dissipated [3,44]. However, the goal is to use materials as much as possible, with technological innovations extending their use [40], aligning with sustainable development objectives.

Although the circular economy is centred on reusing and recycling, energy cannot be recycled, making a genuinely cyclic system unattainable. The idea of one single cyclic system is unrealistic [44]. Based on the second law of thermodynamics, "recycling will always require energy and will always be incompletely generating wastes and side-products (increasing entropy, decreasing exergy) of its own" [3]. Although achieving complete recycling using infinite solar energy is theoretically possible, it is unrealistic [3].

The economic and population growth in urban areas has placed an essential issue in the hands of public managers: how to treat the growing urban waste, including electronic waste [12]. In most cities, the current answer to this question is to send waste to landfills [16]. This strategy, however, can cause significant damage to the soil and water bodies around the landfills and the emission of greenhouse gases.

Understanding how the circular economy addresses this issue involves closely examining waste management concepts. Among these, handling electrical and electronic equipment, e-waste, solid waste,

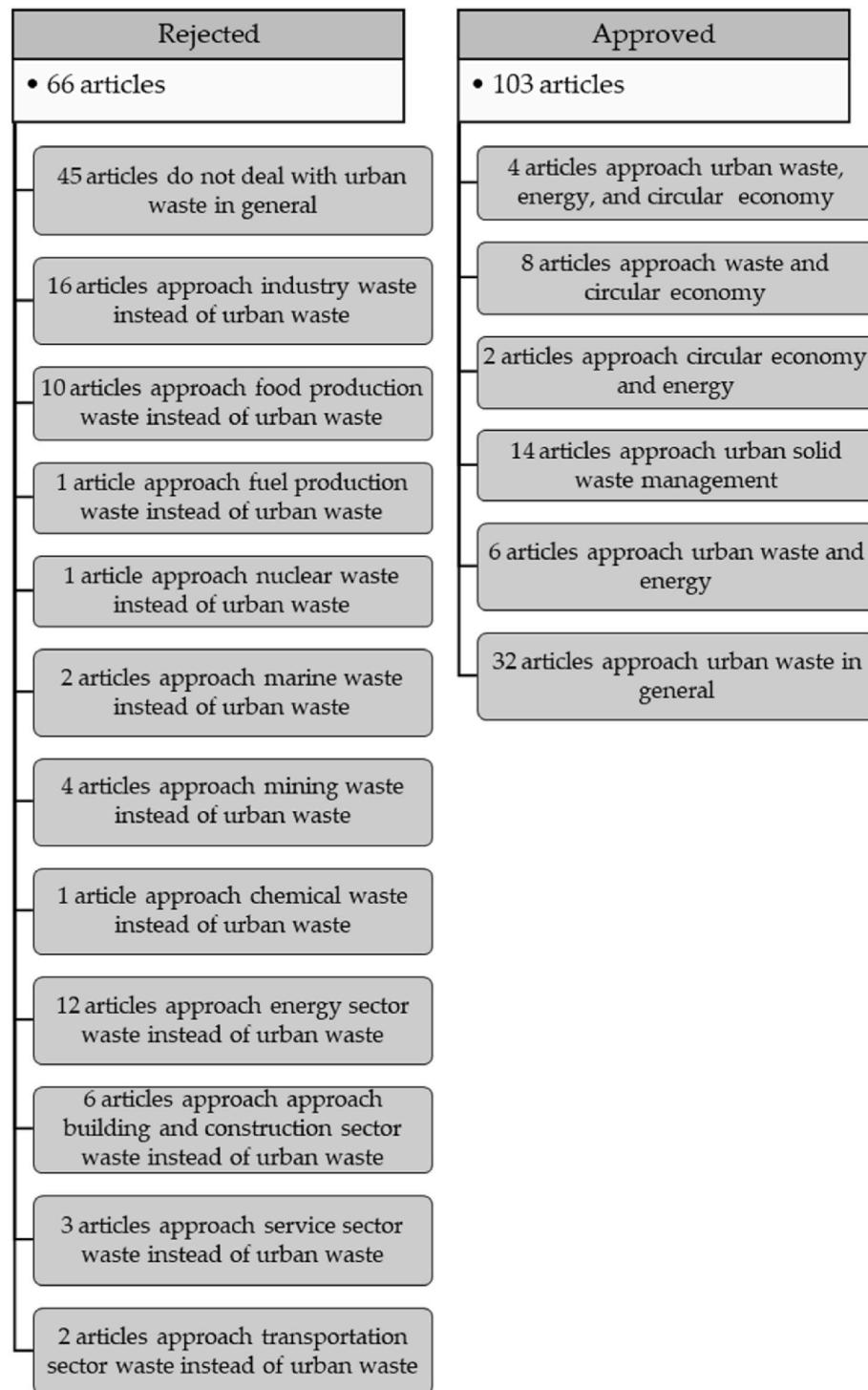


Fig. 5. Filter 2 details and results. Source: Elaborated by authors.

recycling, and reverse logistics takes precedence.

The E.U.'s advances in waste treatment legislation date back to 1975 with the Waste Directive [13]. Nevertheless, the guidelines are periodically improved, with emphasis on the Green Paper on Integrated Product Policy (2001), which introduced the life cycle thinking principle to waste legislation, the Eco-design Framework Directive (2009/125), and the Waste Framework Directive (2008/98) [13]. In addition, with the publication of The Circular Economy Package in 2015, the E.U. established specific regulations for the circular economy. More recently, C.E. and USWM were featured in The European Green Deal, tabled by

the European Commission in 2020 [4]. Due to this, the E.U. legislation is the most advanced in creating a circular economy guideline [44], and the E.U. is leading the way towards enhancing a circular economy [4].

Countries have drawn national policies to promote a circular economy on urban waste management [5]. Specifically, China has incorporated WtE power plants into its national plan to increase renewable energy, as outlined in the 13th Five-Year Plan for Renewable Energy Development. This plan mandates that local governments support the construction and operation of WtE facilities. The initiative reflects China's strong governmental support, backed by subsidies and tax

incentives, ensuring that WtE projects align with national environmental and economic goals. Specifically, the National Policy for Solid Waste was instituted in Brazil in 2010 to promote more responsible and sustainable solid waste management. More recently, the country launched the ProteGEEr project, a partnership with the German government to improve urban solid waste management. As a result, the Brazilian Ministry for Regional Development published official documents guiding the public, especially public managers, on how to address this issue.

Also, Brazil recently implemented the Renovabio Programme, which aims to decarbonise the national road fleet by stimulating the biofuel market in the country through a market-oriented strategy [20]. This policy recognises the importance of all biofuels in the Brazilian energy matrix, including biomethane, and thus aims to promote their development. Considering biomethane generation because of urban solid waste treatment and its applicability as a vehicular natural gas (VNG) substitute, the WtE technology could benefit from the Renovabio and The Future Fuel Law [21]. Although the program mentions urban waste, the focus is on agricultural waste and relations with agribusiness, given the substantial sector's power in Brazilian politics and economy.

3.1. Results: WtE and PESTEL model

In this section, divided by the PESTEL perspectives framework, we will build a strategic analysis of the WtE industries. The political, economic, social, technological, legal, and Environmental dimensions will be considered from a comparative perspective with the selected countries or regions.

3.2. Economic factors

The WtE industry, which involves converting waste materials into useable energy, influences various economic factors in Brazil, the USA, China, India, and the E.U. These factors include inflation, interest, economic growth, and exchange rates, each playing a critical role in shaping the industry's performance and decision-making in these regions.

The cases of the USA and the E.U. are similar due to their high-income status, which strongly influences the investment and development of new technologies in this industry. The well-developed E.U., supported by strong economic growth, low inflation, and favourable interest rates, encourages investment in sustainable technologies. The E.U.'s stable economic environment and robust regulatory framework provide a conducive atmosphere for the WtE sector. Additionally, favourable exchange rates facilitate the import of cutting-edge technologies and foster competitive pricing in the global market. Economic incentives and stringent waste management regulations further bolster the industry's growth, making the E.U. a leader in WtE innovation and implementation [18].

The WtE industry in the United States, despite being a mature and globally proven technology, faces a unique set of challenges compared to its counterparts elsewhere. The primary obstacle remains the historical prevalence of inexpensive landfill options. These provide significantly lower tipping fees than those required to make new WtE facilities economically viable. This cost disparity and a societal preference for "out of sight, out of mind" waste disposal have historically stifled investment in advanced WtE infrastructure.

US energy policy, however, has not been entirely devoid of support for sustainable waste management solutions, albeit with a more targeted approach. The Environmental Protection Agency's (EPA) Landfill Methane Outreach Program (LMOP) is a prime example. This voluntary initiative actively encourages the capture and beneficial use of methane generated in landfills, a potent greenhouse gas. While not directly promoting new WtE incineration facilities, this programme demonstrates a policy inclination towards mitigating immediate environmental concerns by leveraging existing waste infrastructure, rather than embarking

on extensive new WtE developments.

A more robust and comprehensive policy framework is essential for the WtE sector to expand its footprint in the US. Although recent legislation like the Inflation Reduction Act (IRA) offers some potential tax credits for clean energy projects, the industry needs clearer, more direct incentives to overcome the ingrained economic advantage of landfills. A carbon price or higher landfill taxes could help level the playing field. Furthermore, greater inclusion of WtE within renewable energy portfolio standards (RPS) and enhanced public understanding of its multifaceted benefits—including clean energy generation and resource recovery—are crucial for fostering political and financial support.

China's WtE industry is a compelling case study of rapid expansion driven by deliberate policy choices. Fueled by extraordinary economic growth and urbanisation, the Chinese government has substantially invested in WtE infrastructure. This isn't just a reaction to the sheer volume of waste generated by its massive population; it's a strategic imperative. Despite facing economic realities like moderate inflation and relatively stable interest rates, the WtE sector thrives due to significant state subsidies and robust incentives. Furthermore, the government's active exchange rate management is crucial in controlling costs associated with importing advanced WtE technologies, bolstering the industry's growth and technological sophistication. While global market fluctuations and potential trade tensions could pose challenges, China's proactive governmental support has largely shielded the sector, ensuring its rapid and sustained development.

In India, the WtE industry is nascent and faces numerous economic hurdles. High inflation rates and variable interest rates make financing projects challenging. Despite robust economic growth, the sector needs more infrastructure and more investment. Additionally, the weak Indian Rupee increases the cost of importing necessary technology and equipment, hindering industry development. Nonetheless, India's increasing urbanisation and waste production provide significant potential for growth in the WtE sector, contingent on improving economic stability and investment conditions.

The industry is emerging in Brazil but faces economic challenges such as high inflation rates and fluctuating interest rates, which can deter investment. However, the panel data analysis indicates that Brazilian interest rates have converged with those of other countries more due to the crises caused by the pandemic in 2020 and the Ukraine war in 2022 than due to the control of public debt (Fig. 6). Controlling inflation, however, comes at the cost of maintaining high interest rates for private investment. This barrier leaves the market reliant on public policies with direct state investments, subsidies, or differentiated credit rates. The country's economic growth has been inconsistent, impacting long-term planning and infrastructure development. Moreover, exchange rate volatility affects the cost of importing technology and equipment, making it more expensive for the industry to advance. However, Brazil's highest relative urban population among the selected countries presents significant opportunities for the WtE sector, which is connected to public policies.

3.3. Social

Social factors such as demographic changes, cultural norms, and consumer attitudes significantly influence the development and acceptance of the WtE industry in the selected countries. Each region's unique societal trends shape the market demand and organisational practices within the WtE sector, impacting its growth and sustainability [16].

The E.U. showcases a highly developed WtE industry driven by a solid societal commitment to environmental sustainability and waste reduction. Demographic trends, such as urbanisation and an ageing population, influence waste generation patterns and demand for efficient waste management solutions. Cultural norms strongly support recycling and sustainable practices, fostering a favourable environment for WtE technologies. Consumer attitudes are mainly positive, with high public awareness and acceptance of WtE solutions bolstered by stringent

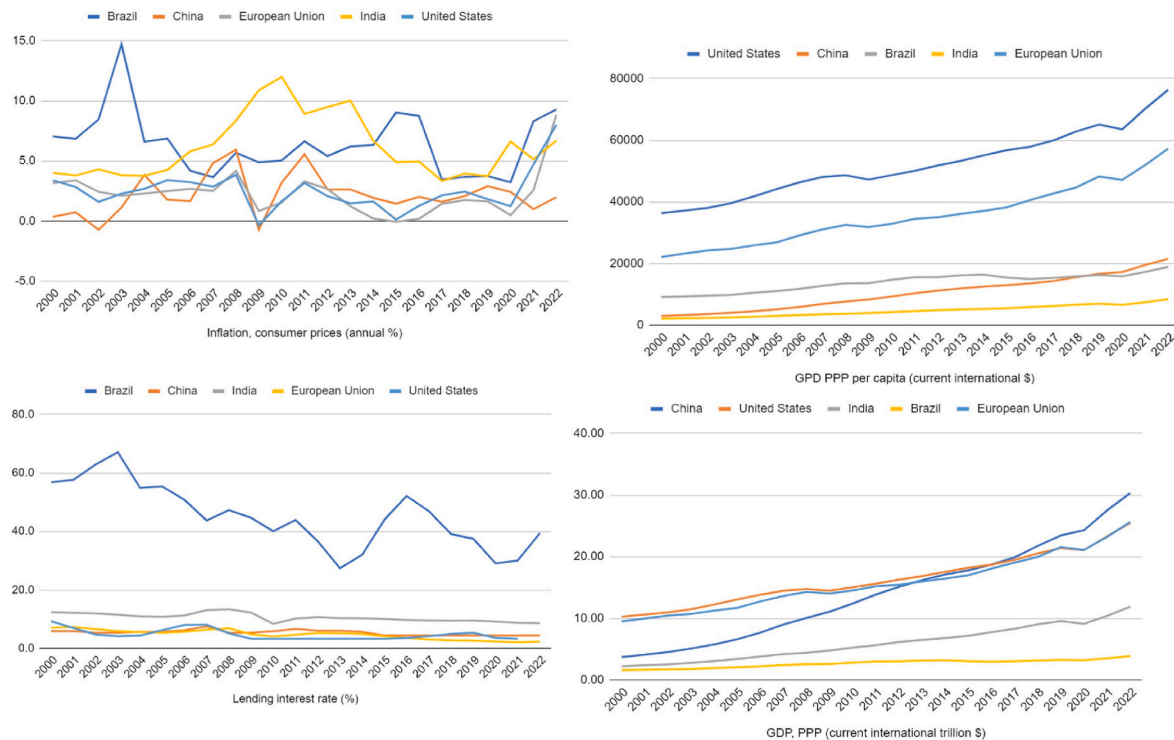


Fig. 6. Inflation current prices, annual% %; GDP PPP per capita, current international; Lending interest rate (%); GDP PPP, current international, trillion \$. Sources: [41–43].

environmental regulations and policies promoting a circular economy [3,6]. In the USA, societal trends favour technological innovation and environmental sustainability, influenced by a strong culture of consumerism and waste generation. The increasing public concern for sustainable practices and governmental support for green technologies boost the WtE industry. However, regional differences in attitudes towards waste management and energy production can result in uneven industry development across the country [12,16].

China's rapid economic growth, unprecedented urbanisation and industrialisation have created immense challenges in waste management. This demographic shift has resulted in the generation of vast quantities of municipal solid waste. While the historical emphasis on rapid economic development often overshadowed environmental considerations, there has been a significant shift in recent years. The Chinese government has increasingly prioritised sustainability and environmental protection, directly influencing the trajectory and regulation of the WtE industry. A key aspect of this shift is the stringent environmental controls now applied to WtE facilities nationwide. China's WtE power plants adhere to pollution control standards comparable to those in the European Union and the United States. This commitment is evident in implementing advanced emission control technologies and rigorous monitoring protocols, ensuring that environmental impacts are minimised despite the sector's rapid expansion. Consumer attitudes are also evolving, increasingly supporting innovative waste management solutions, driven mainly by governmental policies promoting environmental consciousness. Despite these advancements, the widespread success and efficiency of WtE initiatives in China still face hurdles related to public awareness and participation. While government policies encourage environmental responsibility, public engagement in critical areas like waste segregation and recycling remains a developing process. The practical implementation of WtE relies heavily on source separation of waste, and inconsistent public participation in these efforts can impact the overall efficiency and acceptance of WtE projects. Continued efforts in public education and community involvement will be crucial to fully leverage the potential of

China's WtE strategy.

In India, accelerated demographic changes, such as a growing urban population and an increasing middle class, contribute to higher waste generation and demand for effective waste management solutions. Cultural norms around waste disposal are gradually shifting, with rising awareness of environmental issues. However, the societal emphasis on informal waste collection and recycling sectors presents challenges and opportunities for the WtE industry. Consumer attitudes towards WtE technologies are mixed, with acceptance growing in urban areas but limited in rural regions due to a need for more awareness and infrastructure [44].

In Brazil, social factors such as high urbanisation, demographic stabilisation, and rising awareness of environmental issues are driving the WtE industry. Cultural norms around waste segregation and recycling are still developing, posing challenges for the WtE industry. Public acceptance and understanding of WtE technologies still need to be improved, which can affect market demand, policy decisions, investment priority and the pace of industry growth [45].

An important factor for socio-economic development, WtE presents an opportunity for job creation that is unevenly utilised among the selected countries (Fig. 7 and Table 1). Panel data analysis demonstrates that Brazil and the USA could generate significantly more jobs from biogas, municipal and industrial waste, and biomass. The E.U. accounts for nearly half of the WtE jobs from municipal and industrial waste, while China holds a similar proportion of WtE jobs from biogas. Brazil stands out in liquid biofuel employment due to its agricultural sector and photovoltaic industry, although it has considerable potential for WtE in the urban context.

3.4. Technological factors

The panel data also reveals significant disparities in the application of waste management techniques and their utilisation in WtE processes. The E.U., as the largest biogas producer, has around 400 plants, not counting incineration plants. This industry is supported by almost 100 %

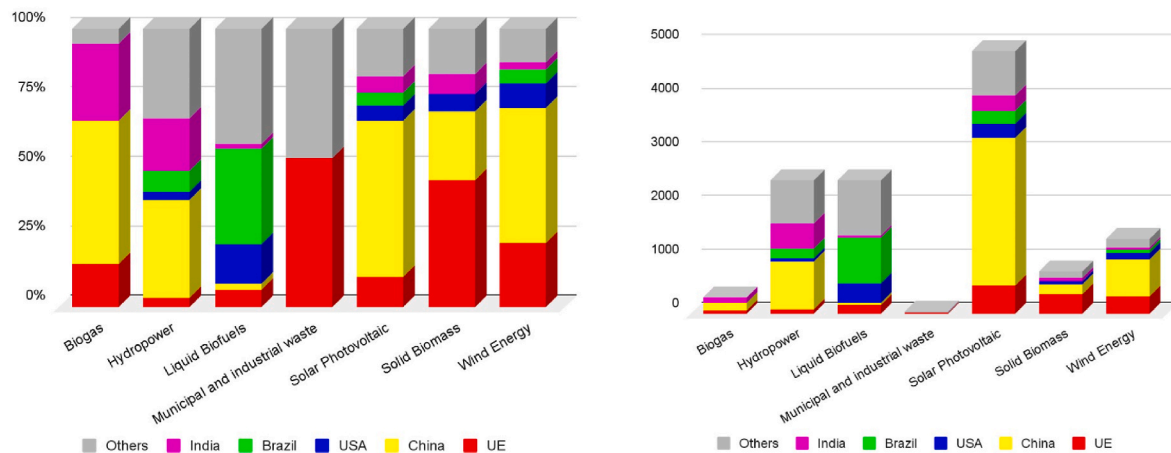


Fig. 7. Jobs by energy Source (thousand). N.D. means “no data”. Source [47].

Table 1

Jobs by energy Source (thousand). N.D. means “no data” available in the source.

Thousands of jobs	Biogas	Hydropower	Liquid Biofuels	Municipal and industrial waste	Solar Photovoltaic	Solid Biomass	Wind Energy	Total
UE	47.1	82.873	148.3	14.5	517.159	353.8	319.26	1482.992
China	160	876.032	55	N.D.	2760	195	681	4727.032
USA	N.D.	66.272	359.709	N.D.	263.883	47.014	125.58	862.458
Brazil	N.D.	193.569	856.15	N.D.	241.443	N.D.	67.724	1358.886
India	85	465.997	35.055	N.D.	282	58	40	966.052
Others	16.695	800.407	1036.065	12.554	837.861	125.442	166.782	2995.806
World	308.795	2485.15	2490.279	27.054	4902.346	779.256	1400.346	12393.226

Source [46].

waste collection and treatment through the biogas, recycling, and incineration industries. Landfill treatment is reduced to practically zero in Northern and Central E.U. countries, with higher rates than in the USA. Brazil has immense potential and needs more infrastructure and investment in city waste treatment. As a purchasing advantage, in addition to being the fifth largest producer of garbage, the country has higher urbanisation rates and organic garbage composition than the USA and the EU. These factors are essential for biogas production due to the possibility of economies of scale and the irrelevance of gas or energy transportation costs. Most waste remains in landfills (76.1 %), and around 16 % in open dumps without treatment. The scenario is better than in 2013, when 44 % of the population lived in urban areas, and around 42 % of the country’s urban waste received inadequate treatment [12,16]. Incineration is not used to produce electrical energy, as China and the E.U. are leaders in this regard. The rest is not even collected, incinerated in environmental protection procedures, buried, or dumped in streams and the sea. If the Brazilian situation is not as severe as India’s, the country will still recycle the least among the countries compared (Table 2).

Technological factors significantly influence the Waste-to-Energy (WtE) industry across different regions, with varying rates of innovation, investment, and technological adoption. The USA, with its robust innovation ecosystem, employs a diverse range of WtE technologies. Mass-burn incineration is the most common, but continuous upgrades ensure compliance with stringent emission standards. Other technologies, such as gasification, pyrolysis, and anaerobic digestion, are also deployed, catering to specific waste streams and regional needs. This high rate of technological change reflects a continuous drive for greater efficiency and enhanced emissions reduction, supported by collaborations between the public and private sectors [18].

The EU’s technological development in WtE is a fundamental pillar of its environmental and economic strategy, driven by stringent regulations like the Waste Framework Directive. This regulatory push actively disincentivises landfilling, creating a strong market for

Table 2

Data 2022. Some data from India and China are controversial in studies and official statistics.

Sources: [42–44].

Country	China	India	EU	USA	Brazil
Urban Population Share (2020)	61.4 %	34.9 %	80.0 %	82.7 %	87.10 %
Waste Collection Coverage - Population %	94 %	78.2 %	97.2 %	100 %	93 %
Composition Organic Waste %	61 %	25–40 %	34 %	15 %	51 %
Composition Plastic %	10 %	NA	11 %	13 %	13 %
Landfill %	60.1 %	18 %	5–10 %	52.6 %	76.1 %
Incineration %	29.8 %	5 %	25.7 %	12.80 %	0.00 %
Open dump %	8.2 %	77 %	0 %	0 %	15.6 %
Recycling %	5–20 %	5.0 %	28.5 %	34.60 %	4 %
Jobs (Thousands)	10000		2300	861	800

advanced WtE solutions. The EU’s holistic circular economy framework prioritises waste prevention, reuse, and recycling, positioning WtE as a last resort for residual waste. This approach fosters innovation in technologies that efficiently recover energy and materials, with significant advancements in flue gas treatment and the exploration of next-generation processes like plasma arc technology. The collaborative approach to R&D, supported by EU funding, ensures continuous improvement in efficiency and resource recovery, solidifying the EU’s global leadership in sustainable waste management [6].

In China, technological advancements are the driving force behind the WtE industry, propelled by strong governmental support and substantial R&D investment. The country has become a global leader in WtE

implementation, primarily through state-of-the-art mass-burn incineration plants. Moving grate technology is the prevailing choice for its ability to process untreated MSW, handle varied waste compositions, and efficiently remove combustion products. These plants feature sophisticated flue gas treatment systems that ensure emissions meet international standards. China is also exploring other technologies like gasification, pyrolysis, and anaerobic digestion to diversify energy recovery. The nation's focus on technological self-sufficiency and the sheer scale of implementation highlights the strategic importance of the sector in reducing landfill reliance and enhancing renewable energy generation. Ongoing research into carbon capture and AI-driven optimisation further demonstrates China's commitment to innovation [33].

In India, the primary technological challenge is the high moisture and heterogeneous content of its MSW, which makes direct incineration less efficient. Consequently, bio-methanation (anaerobic digestion) has gained significant traction for converting high-moisture organic waste into biogas. While large-scale incineration projects exist, they require extensive pre-treatment to produce refuse-derived fuel (RDF). The rate of technological change is slower due to limited R&D funding, a preference for imported technologies, and infrastructural constraints. Despite these hurdles, WtE is increasingly recognised as a critical component of India's Integrated Solid Waste Management (ISWM) strategy, supported by government initiatives like the Swachh Bharat Abhiyan [33].

Brazil is experiencing gradual technological advancements in its WtE industry, driven by the growing recognition of the need to address urban waste challenges. The country is exploring various technologies, including anaerobic digestion and landfill gas recovery. The rate of change is moderate, with a focus on cost-effective, locally adaptable solutions. Brazil is highly competitive in producing solid and liquid biofuels from its agricultural sector, but its capacity for WtE from municipal waste requires further development to match that of other markets. Overall, the technological factors, including innovation and the rate of change, are pivotal in determining how effectively each of these regions leverages technology for waste management and energy generation, ultimately shaping the industry's growth and sustainability [5, 44]. (Table 3).

Waste management from the perspective of the circular economy and environmental protection also has beneficial economic effects for job creation. According to the U.N. and the International Labour Organisation (ILO), between 15 and 20 million people collect, sort, and recycle waste generated by cities worldwide. Brazil has 800 thousand people working in this area. This amount is higher than the renewable energy sector, with 13.7 million jobs worldwide (Tables 1 and 2). Increasing recycling in the country would increase jobs. However, the currently low installed capacity of Biogas and Municipal and Industrial Waste, compared to other renewable energy sources, undervalues them to the point that the search for market data is unavailable, like other sources.

Europe is today's largest biogas producer. Germany is the largest market, home two-thirds of Europe's biogas plant capacity. Energy crops were the primary choice of feedstock that underpinned the growth of Germany's biogas industry. However, policy has recently shifted more towards using crop residues, sequential crops, livestock waste and the

capture of methane from landfill sites. In China, policies have supported the installation of household-scale digesters in rural areas to increase access to modern energy and clean cooking fuels; these digesters account for around 70 % of installed biogas capacity today. Remuneration via the Clean Development Mechanism (CDM) was a key factor underpinning this growth, particularly between 2007 and 2011. The development of new biogas projects fell sharply after 2011 as the value of emission reduction credits awarded under the CDM dropped. The primary pathway for biogas in the USA has been through landfill gas collection, which accounts for nearly 90 % of its biogas production today. There is also growing interest in biogas production from agricultural waste since domestic livestock markets are responsible for almost one-third of methane emissions. Due to state and federal support, the country is also leading the way in using biomethane globally in the transport sector [49].

Brazil has also supported biogas through auctions, and production comes mainly from landfills, but there is also potential from vinasse, a by-product from the ethanol industry. Around 90 % of today's production is from upgrading biogas. Crop residues provide around half the global biogas potential today, but landfill gas is the lowest-cost source. Biogas is produced and consumed locally, meaning transportation costs are negligible. Upgrading biogas captured from landfill sites is typically the cheapest option to produce cost-competitive biomethane. Electricity and liquid biofuels are the primary vectors for decarbonising transport, but biomethane finds a niche in some countries and sectors. In contrast, Europe, China, and the USA account for 90 % of global production.

In Brazil, there are nine electric power generation plants from urban solid waste, most of which are in the southeast region, and all were implemented between 2004 and 2015. Also, the country has three biomethane generation plants from USW in operation, two in the Rio de Janeiro state and one in the Ceará state [22]. Besides, in the 1980s, the now-closed Caju biomethane plant supplied the Municipal Urban Cleaning Company of the State of Rio de Janeiro's fleet. However, the lack of efficient technologies for biogas treatment at the time blocked the project's development [80]. Beyond USW, two biomethane plants in Brazil direct the coproduct to vehicle fuel – both are agriculture and livestock waste management plants (20). Another highlight is the importance of the U.N.'s Clean Development Mechanism for advancing Brazilian WtE plants. The country has 59 Municipal Solid Waste Management projects working under the CDM guidelines, of which the majority (52 %) focus on generating electricity and biomethane.

The country managed to triple its installed capacity from 2014 to 2022. Nevertheless, the government must catch up to its capacity and market development (Fig. 8 and Table 4).

Electrical energy generation from biogas is mainly divided between distributed micro or mini-generation plants and those registered or authorised by the National Electric Energy Agency (ANEEL). Distributed micro and mini generation are characterised by the electrical energy compensation system, initially regulated by 2012. Under this system, which allows the generation of credits to reduce consumption, the first biogas plants appeared in 2014. By the end of 2022, 104.9 MW of biogas power was installed in 435 generating units, adding up the different materials: raw materials – animal waste, urban waste, agroindustry, and

Table 3
Power installed M.W. and total biofuels production TWh (2021).

Country	China	India	EU	USA	Brazil	Others	World
Renewable municipal waste	11022	217	4094	1270	30	2887	19520
Solid biofuels and renewable waste	0	0	19908	11140	15883	72282	119213
Biogas	1711	14	11957	2310	413	5169	21574
Bioenergy	29753	10592	33969	13574	6300	49183	143371
Bagasse	0	0	151	0	12063	8435	20649
Other solid biofuels	17021	10361	15662	9871	3790	22338	79043
Liquid biofuels	0	0	2105	123	5	351	2584
Total Biofuels Production	41.2	26.9	175.4	451.8	254.0	237.8	1187.2

Source: [47,48].

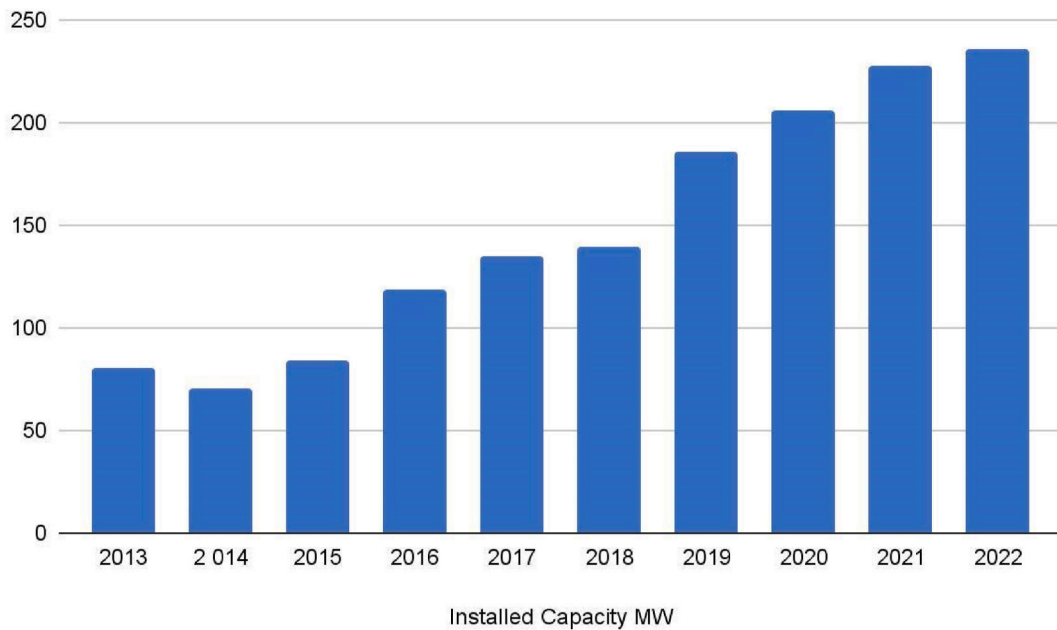


Fig. 8. Biogas installed capacity in M.W.
Source: [50].

Table 4
Biogas installed capacity in M.W.
Source: [50].

Installed Capacity M.W.	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Biogas	80	70	84	119	135	140	186	206	228	236

forestry [90]. The compensation system includes the largest biogas plants, but the rules limit the installed capacity to 5 MW. In 2022, according to ANEEL's Generation Information System, biogas ended the year with 47 plants with registration or authorisation, totalling 236 MW of inspected capacity. Most of these plants, corresponding to 193.1 MW, use biogas from urban solid waste as fuel. That year, three plants operated with a total capacity of 3.1 MW.

The sugar-energy sector is unique for the development of biogas in Brazil due to the high production potential from residues from sugarcane processing and harvesting. Large plants can be installed in sugarcane mills, with different business models for biogas. This solution is due to its intrinsic flexibility and the options available in the context of a plant. One energy use that could bring environmental and financial benefits would be using this source to replace the diesel consumed in this segment. The local production of biomethane and the manufacture of tractors and trucks with this possibility of use would favour the verticalization of the chain, bringing greater competitiveness. Given the importance of carbon reduction certificates for WtE development in Brazil, technology is expected to benefit from the Renovabio Programme.

3.5. Environmental

The panel data analysis reveals that India now emits absolute emissions comparable to those of the E.U., and China emits more than the combined total of the USA, E.U., Brazil, and India. However, per capita

emissions in the USA are nearly equivalent to the combined total of China and the E.U. Over the past decade, China, India, and Brazil have slowed the increase in per capita emissions. In contrast, the E.U. and the USA have reduced these emissions. The scenario is more complex when considering methane emissions from urban waste, livestock farming, and natural gas and coal extraction. Total emissions are not proportional to the population of the analysed regions, indicating the EU's relatively lower environmental impact, potentially due to its lower natural gas production. Brazil and the USA exhibit high per capita methane emissions due to their livestock production combined with the natural gas industry. The demographic pressures in China and India drive emissions related to urban organic waste and animal protein production (Fig. 9). Although approximately 10 % of methane emissions in Brazil and the United States are related to urban waste, utilising this gas in the energy industry could help reduce the presence of methane in the atmosphere, as CO₂ emissions are less harmful. In the case of incineration methods, employing existing technologies for carbon capture and storage is recommended.

In the USA, environmental factors significantly influence the WtE industry. Strict environmental regulations, particularly concerning emissions and waste management, substantially impact WtE facility operations [12]. Moreover, sustainability practices and the impact of climate change are increasingly shaping the industry's direction, with a growing emphasis on cleaner energy technologies [6]. The E.U. strongly emphasises environmental sustainability, which directly affects the WtE industry. Comprehensive ecological regulations, such as the Waste

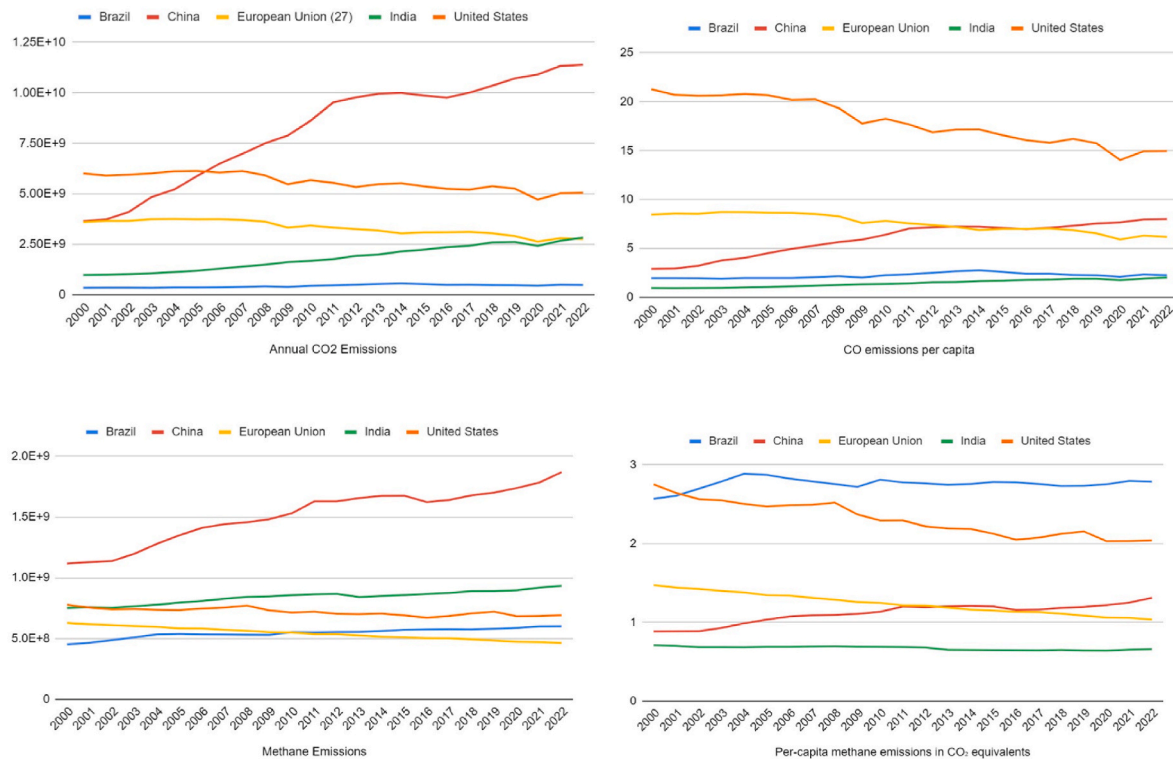


Fig. 9. Annual CO2 Emissions; CO emissions per capita by year; Methane annual emission: Per capita methane emissions in CO2 equivalents. Sources: [42–44].

Framework Directive and the Industrial Emissions Directive, govern waste management and emissions control in WtE plants [3]. The E.U.'s commitment to reducing greenhouse gas emissions drives innovation in cleaner energy production methods, including WtE technologies.

Environmental factors significantly shape the WtE industry in China, where pollution and resource depletion are pressing concerns. Stringent environmental regulations mandate strict emissions standards and waste management practices in WtE facilities. China's focus on transitioning to cleaner energy sources aligns with the development of WtE technologies, contributing to the industry's growth.

Environmental considerations play a crucial role in developing the WtE industry in India. The government has introduced ecological regulations to improve waste management practices and reduce emissions from WtE plants [33]. Furthermore, India's focus on renewable energy and climate mitigation efforts provides opportunities to expand WtE technologies.

Given the country's rich biodiversity and environmental concerns, environmental factors significantly drive Brazil's WtE industry. Environmental regulations dictate waste management practices and emissions control in WtE facilities [42]. Brazil's commitment to reducing carbon emissions and promoting renewable energy creates favourable conditions for WtE technology adoption and innovation [5].

3.6. Legal factors

Legal factors are crucial in shaping the WtE industry, and while less quantifiable than other PESTEL dimensions, they reveal significant differences in approach between regions. The USA's legal framework is a complex mix of federal laws, such as the Resource Conservation and Recovery Act (RCRA) and the Clean Air Act, and varied state-level regulations. States like California and Massachusetts have stricter standards and robust incentives, while overall political fluctuations on climate change create uncertainty for the industry [25]. In China, a strong political environment and legal support are provided through national policies like the Renewable Energy Law (2006) and the Circular

Economy Promotion Law (2008). These laws and specific targets in the 13th Five-Year Plan for Renewable Energy Development ensure consistent growth by mandating local governmental support for WtE facilities and providing subsidies [38].

India's legal landscape for WtE is governed by national and state regulations, including the Solid Waste Management Rules (2016) and the National Policy on Biofuels (2018), which offer a framework for waste management and financial incentives [27]. In the EU, the legal environment is defined by stringent, integrated regulations aimed at a circular economy and climate neutrality. The Waste Framework Directive and the Industrial Emissions Directive set ambitious targets for waste management and impose strict emission limits on WtE plants. The Circular Economy Package and its Action Plan further promote a life-cycle approach, setting ambitious landfill reduction targets and encouraging resource recovery [43]. A key element is Extended Producer Responsibility (EPR), which shifts the burden of product disposal to producers, a fundamental change from the linear economy model [13].

This EU approach, however, faces challenges in achieving policy coherence, especially regarding material-based aspects in product design. There is also the need to find the right mix of regulatory instruments and manage uncertainty with eco-innovation. The EU's strategy is driven by both environmental goals and an economic focus on improving energy efficiency and security, aiming to boost markets for secondary materials and enhance competitiveness [4]. Within this framework, biogas and biomethane are considered effective tools for meeting renewable energy targets, though they are secondary to waste prevention and material recovery in the waste hierarchy [43]. The EU's focus on avoiding overcapacity of WtE plants, to prevent the risk of "stranded assets", is a notable point of concern [18].

Brazil's legal framework for WtE is grounded in the National Solid Waste Policy (NPSW) of 2010, which promotes a systemic, integrated approach to solid waste management and expands responsibility across the production chain [12]. The NPSW prioritises environmental protection, public health, and social principles, including the recognition of

recyclable waste as a valuable economic and social good. It sets clear goals for waste reduction, reuse, recycling, and treatment, and encourages institutional collaboration and economic incentives for sustainable practices. The ProteGEEr project, for example, translates these policies into practical actions, using biogas technology from urban solid waste management to address current environmental challenges and advance academic knowledge [11].

While the EU's legal framework for WtE is guided by a focus on innovation, secondary product development, and climate change mitigation, Brazil's legislation has a more general approach based on environmental issues, health concerns, and social principles. Both frameworks adhere to a waste hierarchy that prioritises waste prevention and material recovery over energy recovery. However, Brazil's legal and institutional advancements, such as the NPSW, demonstrate a clear commitment to leveraging WtE as a component of sustainable waste management.

3.7. Political factors

Political factors and public policy are crucial for the WtE industry, with each region showing a distinct approach. In the USA, a mix of federal and state-level policies like the Clean Air Act and Renewable Portfolio Standards provide a regulatory framework, but political debates on climate change can create uncertainty [25]. In contrast, China's political environment is marked by strong governmental support, with national plans mandating the expansion of WtE and consistent policy implementation driven by political stability [38].

India leverages national policies like the Swachh Bharat Mission and the National Policy on Biofuels to provide financial incentives and set renewable energy targets, although bureaucratic hurdles and varying state regulations can impact growth [27, 96]. The EU provides a stable and supportive environment through ambitious policies like the European Green Deal, which includes directives for waste management and renewable energy generation, driving WtE adoption and investment [26].

In Brazil, the political landscape for WtE is shaped by national policies, but political and economic instability can hinder sustained investment. Recent years have seen significant advancements in biogas and biomethane regulation at the federal level, including the establishment of new standards, the recognition of biomethane as equivalent to natural gas, and the creation of a national strategy to encourage its use and reduce methane emissions [38].

These policy developments include the RenovaBio programme, which provides economic and financial incentives for biofuels like biomethane, aiming to meet Brazil's Paris Agreement commitments. Despite its successes, a need remains for specific legislation to define use parameters and allow for the widespread availability of biogas for energy production, as current regulations are exclusive to biomethane from organic waste [20]. Tax incentives like the REIDI regime and access to advantageous financing through institutions such as the BNDES further support the sector's growth.

The upcoming Future Fuel Law is set to revolutionise Brazil's biogas and biomethane sector. It will introduce a mandatory biomethane blend with natural gas, starting at 1 % in 2026, and a mandate for Sustainable Aviation Fuel. This legislation, along with new financial incentives, is designed to significantly boost the presence of renewable fuels in the national energy mix, promoting a more sustainable energy transition and helping Brazil meet its decarbonisation targets. 5. Discussions on the Brazilian case.

The National Policy for Biofuels, also known as the Renovabio Programme, was instituted in 2017 as a strategy to help achieve the Nationally Determined Contributions defined by Brazil in the Paris Agreement in 2015. The policy aims to decarbonise the Brazilian road fleet by stimulating the biofuel market in the country through a market-oriented strategy. Thereby, the policy is conducted according to three instruments: 1) annual decarbonisation targets directed to the fuel

sector; 2) the biofuel emissions certificate, and 3) the Decarbonisation Credits (CBio).

This framework determines the Brazilian national decarbonisation target, and specific targets are directed to different sectors. These targets are obligations to the fuel sector, which can only be achieved by purchasing decarbonisation credits (CBio). Thus, fossil fuel distributors are mandatory participants in the CBio market.

The CBio emission process, on the other hand, begins when biofuel producers or importers request the issuance of the Biofuel Certificate, through which the biofuel's Life cycle carbon is evaluated, including GHG emissions criteria (19). This analysis is performed according to the RenovaCalcMD methodology. It considers the emissions throughout the production process, including land use, land-use change and forestry, and the industrial stage. The examination also covers "products' and coproducts' efficiency, fossil fuel combustion, and electricity consumption". The criteria apply to biofuels produced nationally and abroad [21].

If approved, the inspection firm emits the Certificate of Efficient Biofuel Production, which mainly informs the GHG emissions mitigated by using the analysed biofuel instead of its fossil fuel substitute; in other words, it presents the carbon intensity difference between these two fuels. Based on this indicator, the writer-certified bank or financial institution emits the Decarbonisation Credit (CBio).

Ultimately, CBio is a financial counterpart to avoided emissions. The credits are traded in organised markets, including auctions, and are managed by B3, the Brazilian stock market company [20].

In the context of the Renovabio programme, Brazilian biomethane producers face three main barriers: scale, infrastructure, and application [20]. While most small rural pig and cattle farms lack the necessary scale to afford certification costs, large urban waste landfills and sugarcane-based energy companies have sufficient scale to do so. Regarding infrastructure, vehicle biomethane faces the same challenges as vehicular natural gas: a shortage of filling stations and a small fleet of compatible vehicles. Finally, since it relies on commercialisation, the Renovabio programme does not account for self-consumed biomethane, which is common in the agriculture and livestock sectors [20].

Biomethane generation from urban solid waste has the highest potential to overcome these issues [11]. This is because it has a sufficient scale to profit from CBio commercialisation, benefits from an already established vehicular natural gas market in many Brazilian cities and does not face the self-consumption problem [21]. To promote the sector, a minimum quota for natural gas distribution companies is suggested, similar to policies for ethanol and biodiesel [19]. Implementing more specific regulations to reduce initial investment costs is also crucial [21]. An added advantage of these policies is that biomethane production would be close to the consumer market, helping to mitigate Brazil's insufficient gas pipeline network [20].

4. Conclusions

This study comprehensively assessed the macro-environmental factors influencing Waste-to-Energy (WtE) in Brazil through a comparative analysis with the USA, China, the EU, and India. The findings highlight a significant disparity in WtE infrastructure, with Brazil currently lacking any municipal solid waste (MSW) combustion plants, instead relying on landfill methane capture. This contrasts sharply with China and the EU, where combustion of post-recycling MSW is a primary waste processing route. While Brazil's immense urban population and waste generation potential remain largely untapped for WtE from MSW, the nation demonstrates a competitive edge in leveraging agricultural waste for electricity and biofuel production, effectively competing with the EU and USA in this sector.

Public policies across these regions are a pivotal determinant of WtE development. The EU leads with stringent environmental regulations and robust investment incentives, while China's rapid expansion is driven by strong governmental mandates. In contrast, Brazil's progress

in implementing WtE strategies for urban waste lags behind these regions despite national policies like Renovabio that promote biogas from agricultural sources. Economically, while high-income regions benefit from stable economies and substantial investment, Brazil, as an emerging market, must navigate challenges like inflation. However, its large urban population and existing infrastructure for landfill methane capture present unique opportunities. Socially, Brazil faces the hurdle of establishing a strong recycling culture, but the WtE sector offers significant job creation potential.

Brazil can draw actionable insights from international experiences to advance its WtE initiatives. Key recommendations include refining existing public policies, such as Renovabio, to integrate waste management with gas blending, establishing minimum biomethane production quotas, and introducing targeted policies to reduce the high initial investment costs for WtE projects. The forthcoming 'Future Fuel Law' presents a timely opportunity to explicitly link sustainable waste management with energy generation. By strategically leveraging its potential in urban waste and building upon its competitive agricultural waste sector, Brazil can transition towards a more circular economy and address its significant urban waste challenges.

CRedit authorship contribution statement

Marcos Eduardo Melo dos Santos: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Brenda Silveira:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Hugo Santos:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Fernando Joaquim Ferreira Maia:** Writing – review & editing, Supervision, Resources, Formal analysis, Data curation. **Ana Paula Basso:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Hirdan Costa:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Funding

This research was funded by ANP/FINEP/FUSP/USP, grant number 0443/19, and RCGI/FAPESP, grant number 2020/15230-5. Funded by resources from the investment of oil companies qualified in the R, D&I clauses from ANP Resolution number 50/2015 (PRH 33.1 – Related to Call number 1/2018/PRH-ANP; Grant FINEP/FUSP/USP Ref. 0443/19 and Fapesp Proc. 2020/15230- 5), supported by Shell.

This research was partially sponsored by CAPES grant #88887.006192/2024-00 and INCT of Intelligent Communications Networks and the Internet of Things (ICoIoT) funded by CNPq (proc. 405940/2022-0) and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) Finance Code 88887.954253/2024-00.

We would like to thank CNPq and CAPES for their support (Project number PVH13333-2020. Grant ref. n. 03/2020-PROPEQ-UFPB), as well as the Federal University of Paraíba (Process: 23074.085232/2023-25).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Edmilson Moutinho dos Santos, Hirdan Katarina de Medeiros Costa and Marcos Eduardo Melo dos Santos reports financial support was provided by State of São Paulo Research Foundation and Brazilian National Agency for Petroleum, Natural Gas and Biofuels. Jewellord Nem Singh reports financial support and administrative support were provided by Wilson Center (Washington DC), Erasmus University of Rotterdam, European Research Council. Edmilson Moutinho dos Santos, Hirdan Katarina de Medeiros Costa and Marcos Eduardo Melo dos Santos

reports financial support and administrative support were provided by University of São Paulo. Rui Castro reports financial support was provided by Foundation for Science and Technology. Marcos Eduardo Melo dos Santos, Jewellord Nem Singh, Rui Castro, Eliza Massi, Hugo Santos, Hirdan Katarina de Medeiros Costa and Edmilson Moutinho dos Santos declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the support from the National Agency for Petroleum, Natural Gas and Biofuels Human Resources Program (PRH-ANP), funded by resources from the investment of oil companies qualified in the R, D&I clauses from ANP Resolution number 50/2015 (PRH 33.1 – Related to Call number 1/2018/PRH-ANP; Grant FINEP/FUSP/USP Ref. 0443/19).

We are also grateful to the “Research Center for Gas Innovation - RCGI” (Fapesp Proc. 2020/15230-5), supported by FAPESP and Shell, organised by the University of São Paulo, and the strategic importance of the support granted by the ANP (National Agency of Petroleum, Natural Gas and Biofuels of Brazil) through the R&D clause.

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